DS4SE Spring ’26: General Feedback on Term Projects

Muhammad Yaseen Khan

May 10, 2026

Summary: First of all, the class has shown a commendable effort in applying Software Engineering and Machine Learning concepts to practical problems. The projects demonstrate good understanding, creativity, and analytical thinking. Though all of you have utilised Generative AI tools in your work, no marks will be deducted in this regard. Potentially, I dedcuted marks in the viva componet where the entire team went blank on the asked question. However, I expect that with further improvements in experimentation, documentation, and interpretation of results, these term papers have the potential to reach a much higher standard of academic and technical excellence. Keep refining your work and focus on strengthening the quality and depth of your analysis.

As discussed during the viva, your work requires several important improvements and refinements. **The individual feedback are alaredy provided during demos**, inludign those, the following points are general feedback to all projects. Please carefully address the following points in your report, all of them are maintained in a generalized structure in the subsequent paragraphs.

1 Final Submission

- You have to submit all of your codebase, (data: through shareable link), figures, excel sheets, term paper in an *editable* format.
- This is so because I will add some of the experiments myself and supply SLR on it.
- I will modifiy your papers (and send to the conferences as per the final position of work)
- With this, I expect you have no conflict in publishing our work in the academic venues. If you have any confilct, please write an email while the submission of the project.

2 Abstract

The abstract should present a concise yet complete overview of the research work. It must include the following components in a structured manner:

- 2 lines explaining the problem being addressed in the project/study.
- 2 lines discussing why the problem is important and relevant.
- 2 lines describing the dataset used in the research.
- 2 lines highlighting the issues or limitations present in the dataset.
- 2 lines explaining the steps taken to resolve or improve the dataset quality.
- 3–5 lines describing the proposed methodology and experimental setup.
- 3–5 lines summarizing the outcomes and findings of the experiments.
- 1 line on a motivational kick that your work is nice and it would help other to mitigate the problem understudy.

3 Figures and Table

- Every table and figure must be numbered and refered in the text.
- Use correct plot type for visualzation. For example, if you are working on a binary/ternary classification problem, don't use bar charts; use pie charts instead.
- make sure the figures are not cluttered. Reduced the radius of circles in scatter plots for PCA or clustering visualizations. Make the lines fine in line plots.
- use print-friendly, vibrant colours.

4 Introduction

The Introduction section should clearly establish the context, motivation, and significance of the research problem. It must address the following points:

- Clearly explain **what** problem is being solved.
- Provide sufficient background regarding the problem domain. For example, if the work is related to software defect prediction, explain what software defect prediction is and why defects occur in software systems.
- Discuss **why** this problem is important in Software Engineering.
- Explain why solving this problem in an automated manner is valuable and necessary.
- Discuss how Data Science and Machine Learning techniques can improve the effectiveness and efficiency of the solution.
- Identify the stakeholders and beneficiaries who may gain advantages from the proposed solution.

5 Dataset and Features

The Dataset and Feature Engineering section should be highly detailed and systematic. Every preprocessing or transformation step performed on the data must be properly documented.

- Include detailed explanations of the dataset collection, preprocessing, cleaning, and feature engineering procedures.
- It is highly recommended to present the data collection and preprocessing pipeline using a flowchart.
- Introduce all features used in the study and explain:
  - Why each feature is relevant to the problem.
  - What contribution each feature makes toward solving the problem.
- Perform statistical analysis such as feature correlation analysis.
- Present the complete correlation matrix as a figure if possible.
- Provide a discussion specifically focusing on the relationship between the target feature and the remaining features.
- If feature importance techniques are used, include the corresponding visualizations and analysis.

- Any feature selection or elimination process must be properly justified with logical reasoning and supporting evidence.
- Present the data distribution through tables and visualizations.
- If feature selection/elimination is performed, compare the original dataset distribution with the modified dataset distribution.
- Data balancing techniques such as SMOTE or Random Under Sampling should also be discussed in this section.
- PCA and clustering experiments, along with their visualizations, may also be included here.
- If clustering is performed:
  - Determine the optimal number of clusters using methods such as the Elbow Method or Silhouette Analysis.
  - Provide a proper cluster analysis.
  - Discuss cluster-wise data distribution with respect to the target class.
  - Include textual observations obtained through visual inspection and interpretation of clusters.

6 Methodology

The Methodology section should explain the complete experimental design and machine learning pipeline adopted in the study.

- Begin by describing the overall Data Science or Machine Learning approach selected for solving the problem.
- Provide details regarding the classifiers or algorithms used for the supervised learning task.
- Discuss:
  - Train-test split ratio
  - Stratification techniques
  - Cross-validation methods
  - Any additional evaluation or validation procedures
- A tabular representation of train-test data distribution can also be included.
- Clearly explain the different strategies or experiments conducted in the study. For example:
  - Experiment 1: Original dataset
  - Experiment 2: Selected features only
  - Experiment 3: Selected features with SMOTE
  - ...
- Each experiment should be discussed in separate paragraphs.
- For every experiment:
  - Explain the experimental setup.
  - Mention whether feature selection or balancing techniques were used.
  - Describe the classifiers applied.
  - Discuss the expected outcomes and motivations behind the experiment.

7 Results and Discussion

The Results and Discussion section should critically evaluate and interpret the outcomes of all experiments.

- Begin the section by discussing the evaluation criteria and performance metrics used in the study.
- Clearly explain whether standard evaluation metrics or weighted evaluation metrics are more appropriate for the nature of the problem.
- Clearly state, if your work is more significant around one class, rather than all classes are important.
- Add comparative result tables covering all experiments and classifier configurations.
- Add comparative figures for the system-level metrics (Accuray, Balanced-Accuracy, F1-Measure, etc.). ROC-AUC curvers can be added here with the lines indiciating different experiments.
- If tables exceed page width, rotate them appropriately for readability.
- Include normalized confusion matrices for:
  - The best-performing classification configuration
  - The worst-performing classification configuration
- Provide detailed interpretation and reasoning for the obtained results.
- Perform qualitative analysis and observation-based discussion wherever possible, especially through careful examination of the test split predictions.

8 Conclusion

The Conclusion section should summarize the complete research work in a concise and meaningful manner.

- Summarize the objectives, methodology, and key findings of the study.
- Discuss the limitations and shortcomings of the current work.
- Suggest possible future directions and improvements that may enhance the performance and quality of the proposed solution.